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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/746,607	06/22/2015	Tae Su JANG	K100BK-057800US	4820

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EXAMINER
CHAN, CANDICE

ART UNIT	PAPER NUMBER
2813	

NOTIFICATION DATE	DELIVERY MODE
09/07/2017	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 14/746,607	Applicant(s) JANG ET AL.	
	Examiner Candice Y. Chan	Art Unit 2813	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE *3* MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on *7/18/2017*.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) *1-19* is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6)☒ Claim(s) *16-19* is/are allowed.

7)☒ Claim(s) *1-11 and 13-15* is/are rejected.

8)☒ Claim(s) *12* is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .

4) ☐ Other: _____.

DETAILED ACTION

This Office action is in response to the arguments filed 18 July 2017. Claims 1-19 are currently pending.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Arguments

Applicant's arguments, see pp. 6-9, filed 18 July 2017, with respect to the rejection(s) of claim(s) 1-19 under 35 U.S.C. 103 have been fully considered and are persuasive. Therefore, the rejection has been withdrawn. However, upon further consideration, a new ground(s) of rejection is made in view of US 8,975,140 B2 to Chung and US 7,872,313 B2 to Jung.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 9-15 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention. Claim 9 recites "a first gate electrode buried in the active region and the device isolation film and provided in a first gate trench." It is unclear whether this limitation refers to

three portions of the first gate electrode (buried in (1) the active region, (2) the device isolation film, and (3) a first gate trench) or whether the recited first gate trench is intended to be within the active region or the device isolation film. For the purposes of examination, the latter is assumed.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

Claim(s) 9-11 and 13 is/are rejected under 35 U.S.C. 102(a)(2) as being anticipated by US 2015/0035022 A1 to Chung (hereinafter "Chung").

The applied reference has a common assignee with the instant application. Based upon the earlier effectively filed date of the reference, it constitutes prior art under 35 U.S.C. 102(a)(2). This rejection under 35 U.S.C. 102(a)(2) might be overcome by: (1) a showing under 37 CFR 1.130(a) that the subject matter disclosed in the reference was obtained directly or indirectly from the inventor or a joint inventor of this application and is thus not prior art in accordance with 35 U.S.C. 102(b)(2)(A); (2) a showing under 37 CFR 1.130(b) of a prior public disclosure under 35 U.S.C. 102(b)(2)(B) if the same invention is not being claimed; or (3) a

statement pursuant to 35 U.S.C.102(b)(2)(C) establishing that, not later than the effective filing date of the claimed invention, the subject matter disclosed in the copending application and the claimed invention were either owned by the same person or subject to an obligation of assignment to the same person or subject to a joint research agreement.

Regarding independent claim 9, Chung (Fig. 6) discloses a semiconductor device comprising: a device isolation film **204** defining an active region **202**; a first gate electrode (portion of **210** in trench) buried in the active region and the device isolation film **204** and provided in a first gate trench (in **202**); and a second gate electrode (portion of **210**; see Col. 4, line 66 - Col. 5, line 8) provided in the first gate trench and located over portions of the first gate electrode that are buried in the active region **202**, and not disposed over portions of the first gate electrode that are buried in the device isolation film **204** between adjacent active regions. Chung (Fig. 9) also reads on claim 9 as follows: a device isolation film **204** defining an active region **202**; a first gate electrode **212a/b** buried in the active region **202** and the device isolation film **204** and provided in a first gate trench (in **202**); and a second gate electrode **216** provided in the first gate trench and located over portions of the first gate electrode that are buried in the active region **202**, and not disposed over portions of the first gate electrode that are buried in the device isolation film **204** between adjacent active regions.

Regarding claim 10, Chung (Fig. 6) discloses the semiconductor device according to claim 9, wherein each of the adjacent active regions includes a storage node junction region **203**.

Regarding claim 11, Chung (Fig. 6) discloses the semiconductor device according to claim 9, wherein the first gate electrode (portion of **210**) is formed at a lower level than a junction region **203** of the adjacent active regions.

Regarding claim 13, Chung (Fig. 6) discloses the semiconductor device according to claim 9, wherein the first gate electrode and the second gate electrode have different work functions (the first and second gate electrodes comprise different materials in both interpretations of Chung, see Col. 4, line 66 - Col. 5, line 8 and Col. 5, lines 44-48), the first gate electrode being provided below the second gate electrode (Figs. 6 or 9).

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 14 and 15 is/are rejected under 35 U.S.C. 103 as being unpatentable over Chung in view of US 2011/0260242 to Jang (hereinafter "Jang").

Regarding claim 14, Chung discloses the semiconductor device according to claim 13, however fails to expressly disclose wherein the second gate electrode has a lower work function than the first gate electrode. Jang (Fig. 2) discloses a semiconductor device including a first and second gate electrode **26, 27** buried in an active region in a gate trench, wherein the second gate electrode has a lower work function than the first gate electrode (¶¶ 0023-24). It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Jang to include gate electrodes as taught by Chung for the purpose of reducing gate induced drain leakage (abstract).

Regarding claim 15, Chung and Jang disclose the semiconductor device according to claim 14, wherein the second gate electrode includes one or more of nitrogen (N), oxygen (O), arsenic (As), aluminum (Al), and hydrogen (H) ions implanted into a conductive material of the first gate electrode (Jang, ¶ 0023).

Claims 1-8 is/are rejected under 35 U.S.C. 103 as being unpatentable over Jang in view of US 7,872,313 B2 to Jung (hereinafter "Jung").

Regarding independent claim 1, Jang (Fig. 2) discloses a semiconductor device comprising: a device isolation film **14** defining an active region; a main gate (disposed in second recess **22** from the left) having first and second gate electrodes **26, 27** buried in the active region.

Jang fails to expressly disclose a passing gate having only a single gate electrode buried in the device isolation film. Jung (Fig. 2) discloses a semiconductor device with an expanded

storage node contact region including a passing gate having only a single gate electrode **182b** buried in device isolation film **160**. It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Jang with a passing gate configuration as taught by Jung for the purpose of expanding the storage node contact region and improving device performance (Jung, Col. 4, lines 7-15).

Regarding claim 2, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 1, Jang discloses wherein the first gate electrode **26** is provided below the second gate electrode **27**, and wherein the first gate electrode has a first work function and the second gate electrode has a second work function, the second work function being lower than the first work function (¶ 0023-24).

Regarding claim 3, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 2, Jang discloses wherein the first gate electrode **26** is disposed at a lower level than a junction region **11** of the active region.

Regarding claim 4, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 2, Jang discloses wherein the second gate electrode **27** is formed at a substantially same level as a junction region **11** of the active region.

Regarding claim 5, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 2, the combination discloses wherein the passing gate and the main gate are both disposed in a same line-type gate trench (see Jang, Fig. 2).

Regarding claim 6, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 2, Jung discloses wherein the passing gate is located between adjacent storage node junction regions (Jung, Fig. 2).

Regarding claim 7, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 6, Jang discloses wherein the passing gate is disposed below the storage node junction regions **11**.

Regarding claim 8, Jang (Fig. 2) and Jung disclose the semiconductor device according to claim 2, Jang discloses wherein the second gate electrode includes one or more of nitrogen (N), oxygen (O), arsenic (As), aluminum (Al), and hydrogen (H) ions implanted into a conductive material of the first gate electrode (¶ 0023).

Allowable Subject Matter

Claim 12 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter: the prior art of record, either singularly or in combination, does not disclose the combination of limitations found in independent claim 1, including the limitations requiring: the second gate electrode is formed at a substantially same level as the junction region of the active regions.

Claims 16-19 are allowed.

The following is a statement of reasons for the indication of allowable subject matter: independent claim 16 recites, *inter alia*, a passing gate having only a third gate electrode provided in a second trench, the third gate electrode having an upper surface provided with the second trench; a dielectric film having first and second portions, the first portion extending into the first trench and having a lower end proximate to the upper surface of the second gate

electrode, the second portion extending into the second trench and having a lower end proximate to the upper surface of the third gate electrode, wherein the lower end of the second portion is at a lower level than the lower end of the first portion. These limitations, in combination with the remaining limitations of the independent claim, are not anticipated or rendered obvious by the prior art. Claims 17-19 depend from independent claim 16.

Conclusion

The following prior art made of record and not relied upon is considered pertinent to applicant's disclosure: US 2012/0286354 A1 to Cho.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Candice Y. Chan whose telephone number is (571)272-9013. The examiner can normally be reached on M-F, 8-4:30PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, William (Billy) Kraig can be reached on (571) 272-8660. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/C. Y. C./
Examiner, Art Unit 2813
22 August 2017

/LAURA MENZ/
Primary Examiner, Art Unit 2813